CONFLUENTIA 2026 · PROBLEM STATEMENT 4 · DRAFT

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CareFlow AI

Human-in-the-loop triage for Indian clinics: reads messages, voice notes and reports, suggests who to see first and why, and leaves every decision to a clinician.

Progress and results so far · 22 October 2026

The problem

Many inputs

Typed messages, voice notes, photos of lab reports and prescriptions arrive faster than a doctor can read them.

Many languages

English, Hindi, Hinglish, Marathi, Bengali, Tamil… often mixed in one message.

High stakes

A heart attack hidden in a long queue waits. Data may be missing, conflicting or even tampered with.

Goal: read everything, say who should be seen first and *why*, flag what is missing or doesn’t add up, and let a clinician decide.

How one case flows

1 · Intake

Consent, patient details, message, voice note, report photo or PDF, vitals.

2 · Into text

Speech-to-text with timestamps, OCR with line positions, language detection, hidden-instruction masking.

3 · Extraction

Small model (Qwen2.5 3B) fills a form with exact quotes; keyword and number parsers check it.

4 · Triage (no AI)

~50 red-flag rules, NEWS2 vital score, critical lab values → P1–P4 with reasons.

5 · Flags & questions

Missing, unclear, conflicting, suspicious; follow-up questions in the patient's language.

6 · Clinician decides

Approve, edit or override (reason required). Shadow agent learns from it.

Safety by design

The model reads, rules decide

Priority comes from explainable rules, never from the language model.

Every fact has a source

Click a reason to see the exact words, report line or second of audio.

Doubt goes up, not down

Vague, conflicting, unreadable or tampered input lifts a case to at least P3 for a human.

The learner can only raise

The shadow agent stays hidden until it graduates, then may raise a priority, never lower it.

What is built

Area	Working today
Inputs	Text, voice notes, report photos and PDFs; English, Hindi, Hinglish (+ regional flagged)
Extraction	Symptoms, denials, vitals, labs, medicines, timeline — each with a source link
Checks	Missing info, conflicts, wrong-patient reports, hidden “ignore the rules” text
Triage	Red flags, NEWS2, critical labs, queue with time-to-target
Human in loop	Approve / edit / override with reason; follow-up questions re-check the case
Privacy	Consent slip, withdrawal code, roles, erasure, tamper-evident audit log
Learning	Shadow agent (contextual bandit) with graduation and blind checks
Evaluation	Built-in page: synthetic + real data, CareFlow vs triage nurse

A small model on a laptop

3B

Qwen2.5 3B Instruct, via Ollama (bonus: < 20B)

8 GB

Apple Silicon Mac, no GPU server, data stays local

~17 s

per case (rules alone: 0.2 s)

Making a 3B model reliable

- Forced JSON shape and a fixed symptom list
- Every label must match the words it quotes
- Copied or invented signals are dropped
- A doubtful danger label still goes to a human
- Keyword scanner runs alongside as a safety net

How we tested it

Set	Cases	Role
Synthetic – development	92	Written by us; used while building and fixing
Synthetic – held-out 2	20	Written before the model fixes, labels hash-locked
Real ED data – KTAS-dev	150	Real visits; misses used to find bugs
Real ED data – KTAS-test	150	Never looked at; run once — the honest number

KTAS: 1,267 adult emergency visits from two Korean hospitals (Moon et al. 2019). An expert panel's level is the correct answer; the triage nurse's own level is the human baseline.

Headline metric: urgent patients (P1/P2) rated less urgent — the dangerous mistake. Diagnosis and outcome columns are never shown to the system.

Result 1 • the small model helps on unseen phrasing

Urgent patients rated too low (held-out set 2)

7/10

Rules only

4/10

3B model + rules

Held-out set 2 (20)	Rules	3B + rules
Exact priority	55%	70%
2+ levels too low	6	2
Weighted kappa	0.38	0.66
Time per case	0.2 s	19 s

Keywords work on phrasings they know and miss new ones (“squeezing feeling in my chest”). The model reads paraphrases and Hinglish; the keywords stay as a backstop.

Result 2 • all 112 synthetic cases

All 112 cases	Rules only	3B, first version	3B, latest
Urgent rated too low	12 of 66	7 of 66	5 of 66
Exact priority	85%	85%	86%
Over-triage	3%	9%	7%
Expected warning signs caught	100%	100%	100%
Hidden-instruction flag (recall)	100%	100%	100%

Covers denials (“no chest pain”), Hinglish and Hindi, report images and PDFs, conflicting vitals and tampered reports. Bengali and Tamil without a translation model are flagged for a human, not understood.

Result 3 • real patients, vs the triage nurse

Urgent patients rated too low

26%

CareFlow · 7 of 27 | Nurse · 37% (10 of 27)

Routine patients sent up a level

44%

CareFlow · 44% | Nurse · 2%

KTAS-test (150 unseen visits)	CareFlow (3B + rules)	Triage nurse
Exact priority · weighted kappa	41% · 0.25	84% · 0.77
2+ levels too low	1	1

What real data taught us

Found	Fix
Unresponsive patients (ACVPU P/U) were not P1	Responsiveness from vitals now gives P1 (V/C gives P2)
No rule for cardiac arrest / after CPR	Added, P1
ED shorthand missed (“Abd pain”, “Pain, chest”)	Clinical vocabulary added
~1 in 5 model labels didn’t match its quote	Labels checked against the quoted words

46% → 12% Urgent patients rated too low on KTAS-dev (rules only), before vs after these fixes. Nurse on the same patients: 27%.

Limits and next steps

Honest limits

- Too much over-triage on real data (44%)
- Voice works in the app but is not in the test sets
- KTAS-test has no level-1 patients
- Synthetic cases are written by us
- Bengali / Tamil need a translation model

Next

- Gemini run as a large-model comparison
- Rerun the 3B model after the last safety fix
- Clinic-set over-triage dial (NEWS2 threshold)
- Record the demo: voice note, report, override
- Pilot with real clinic messages

CareFlow suggests; a clinician decides. Every number here was measured, including the ones that don’t flatter us.